Data Analysis
AMS315

Chapter 9

Fundamentals of Hypothesis Testing: One-Sample Tests

Learning Objectives

In this chapter, you learn:

- The basic principles of hypothesis testing
- How to use hypothesis testing to test a mean or proportion
- The assumptions of each hypothesis-testing procedure, how to evaluate them, and the consequences if they are seriously violated
- How to avoid the pitfalls involved in hypothesis testing
- The ethical issues involved in hypothesis testing

What is a Hypothesis?

- A hypothesis is a claim (assumption) about a population parameter:

  - population mean

Example: The mean monthly cell phone bill of this city is μ = \$42

  - population proportion

Example: The proportion of adults in this city with cell phones is π = 0.68

The Null Hypothesis, H_0

- States the claim or assertion to be tested

Example: The average number of TV sets in U.S. Homes is equal to three ($H_0 : \mu = 3$)

- Is always about a population parameter, not about a sample statistic

$H_0 : \mu = 3$

$H_0 : \bar{X} = 3$

The Null Hypothesis, H_0

(continued)

- Begin with the assumption that the null hypothesis is true
  - Similar to the notion of innocent until proven guilty

- Refers to the status quo
- Always contains “=” , “≤” or “≥” sign
- May or may not be rejected

The Alternative Hypothesis, H_1

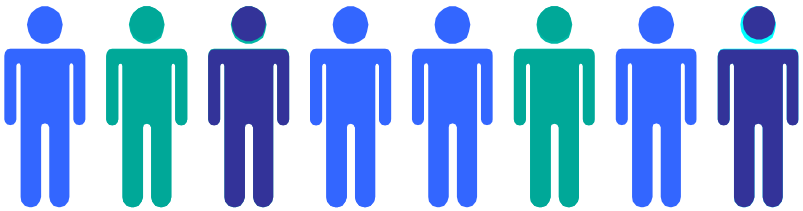
- Is the opposite of the null hypothesis
  - e.g., The average number of TV sets in U.S. homes is not equal to 3 (H_1: $\mu \neq 3$)
- Challenges the status quo
- Never contains the "=" , "≤" or "≥" sign
- May or may not be proven
- Is generally the hypothesis that the researcher is trying to prove (the reason why the research claim is H_1 in single-tail test)

H_0 versus H_1

- If you reject the H_0, you have statistical proof that H_1 is correct.
- If you do not reject H_0, then you failed to prove the H_1.
- The failure to prove H_1 (do not reject H_0) does not mean that you have proven H_1

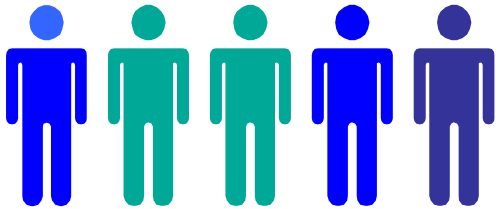
Hypothesis Testing Process

Claim: the population mean age is 50. (Null Hypothesis: H_0: $\mu = 50$)



Population

Now select a random sample



Sample

Suppose the sample mean age is 20: $\bar{X} = 20$

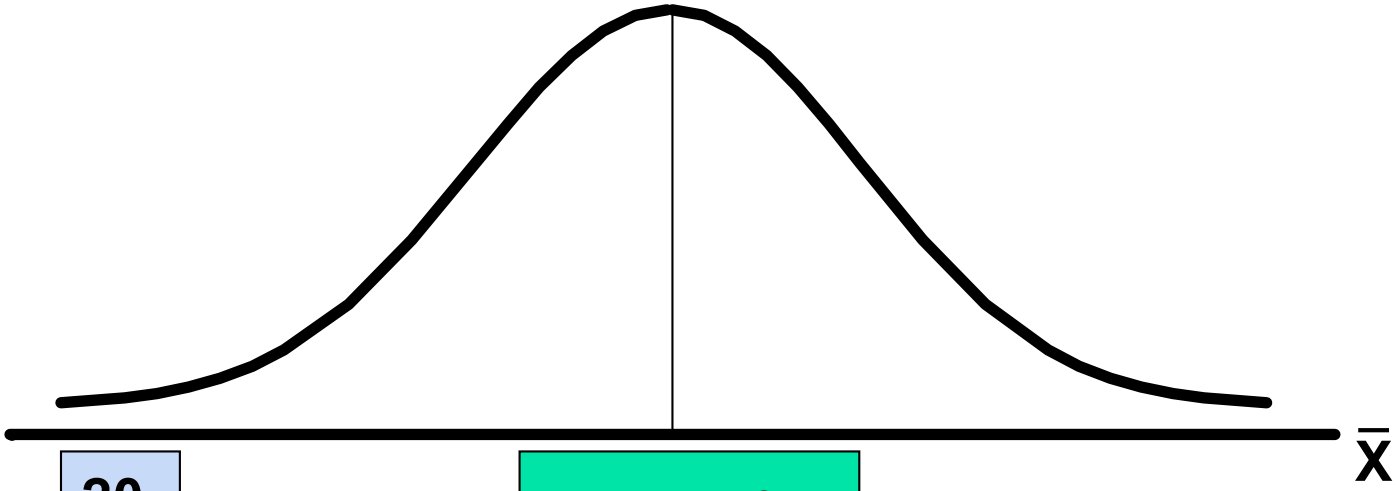
Is $\bar{X}=20$ likely if $\mu = 50$?

If not likely,

REJECT
Null Hypothesis

Reason for Rejecting H_0

Sampling Distribution of $\bar{X}$



If it is unlikely that we would get a sample mean of this value ...

... if in fact this were the population mean…

... then we reject the null hypothesis that μ = 50.

Level of Significance, α

- **Defines the unlikely values of the sample statistic if the null hypothesis is true**
  - Defines rejection region of the sampling distribution
- Is designated by α , (level of significance)
  - Typical values are 0.01, 0.05, or 0.10
- Is selected by the researcher at the beginning
- Provides the critical value(s) of the test

Level of Significance and the Rejection Region

Level of significance = α

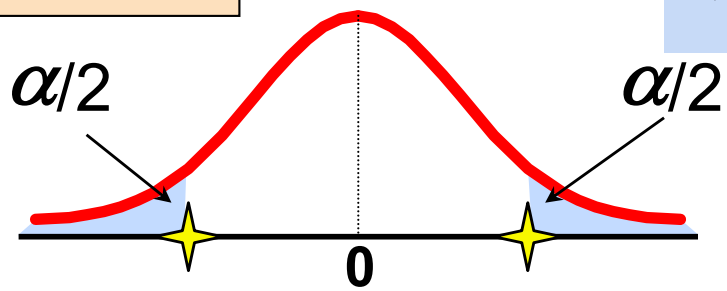
Represents critical value

Rejection region is shaded

H_0: $\mu = 3$

H_1: $\mu \neq 3$

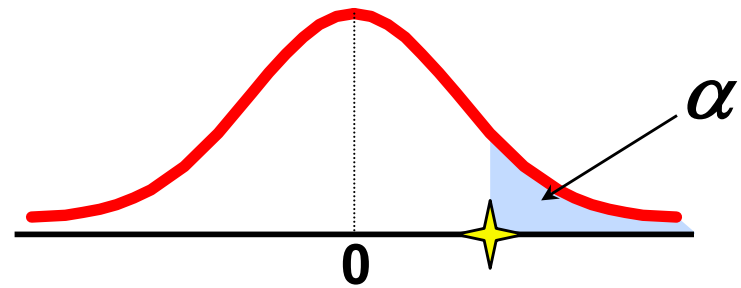
Two-tail test



H_0: $\mu \leq 3$

H_1: $\mu > 3$

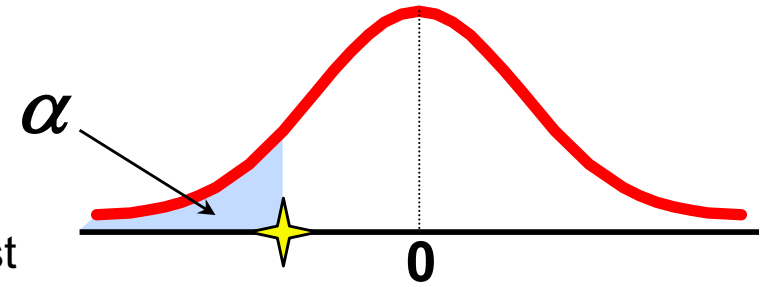
Upper-tail test



H_0: $\mu \geq 3$

H_1: $\mu < 3$

Lower-tail test



Errors in Making Decisions

- **Type I Error**
  - Reject a true null hypothesis
  - Considered a serious type of error

The probability of Type I Error is α

  - Called level of significance of the test
  - Set by the researcher in advance

Errors in Making Decisions

(continued)

- **Type II Error**
  - Fail to reject a false null hypothesis

The probability of Type II Error is β

Outcomes and Probabilities

Possible Hypothesis Test Outcomes

	Actual Situation	
Decision	H_0 True	H_0 False
Do Not Reject **H_0**	**No error** **$(1 - \alpha)$**	**Type II Error** **(β)**
Reject **H_0**	**Type I Error** **(α)**	**No Error** **$(1 - \beta)$**

Key:
Outcome
(Probability)

Type I & II Error Relationship

- Type I and Type II errors cannot happen at the same time
  - Type I error can only occur if H_0 is true
  - Type II error can only occur if H_0 is false

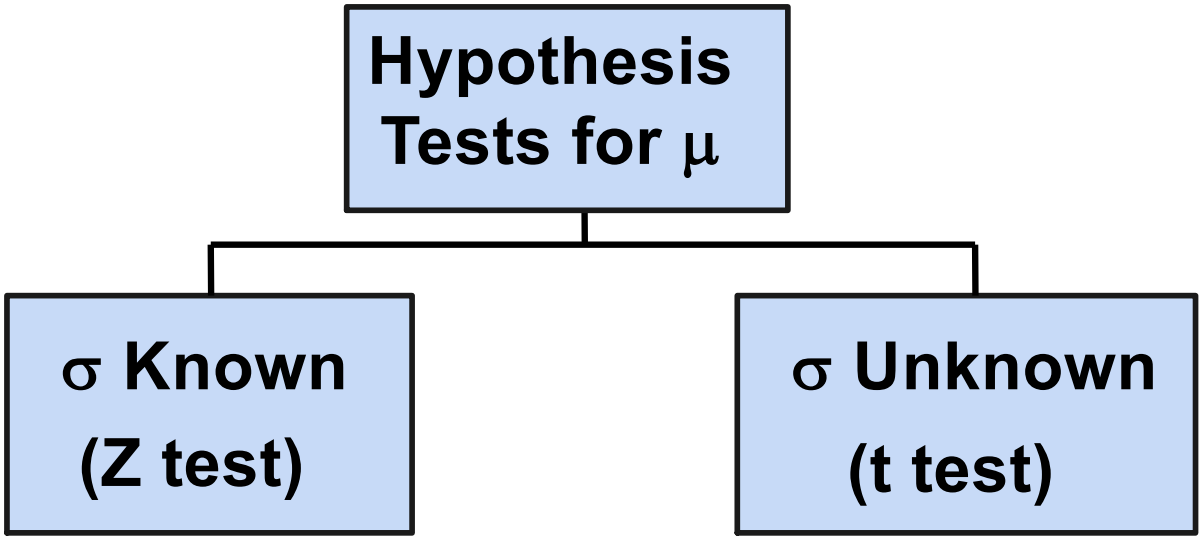
If Type I error probability (α) $\Uparrow$, then

Type II error probability (β) $\Downarrow$

Factors Affecting Type II Error

- All else equal,
  - $\beta \uparrow$ when the difference between hypothesized parameter and its true value $\downarrow$
  - $\beta \uparrow$ when $\alpha \downarrow$
  - $\beta \uparrow$ when $\sigma \uparrow$
  - $\beta \uparrow$ when $n \downarrow$

Hypothesis Tests for the Mean



Z Test of Hypothesis for the Mean (σ Known)

- Convert sample statistic ($\bar{X}$) to a Z test statistic

Hypothesis Tests for μ

σ Known (Z test)

σ Unknown (t test)

The test statistic is:

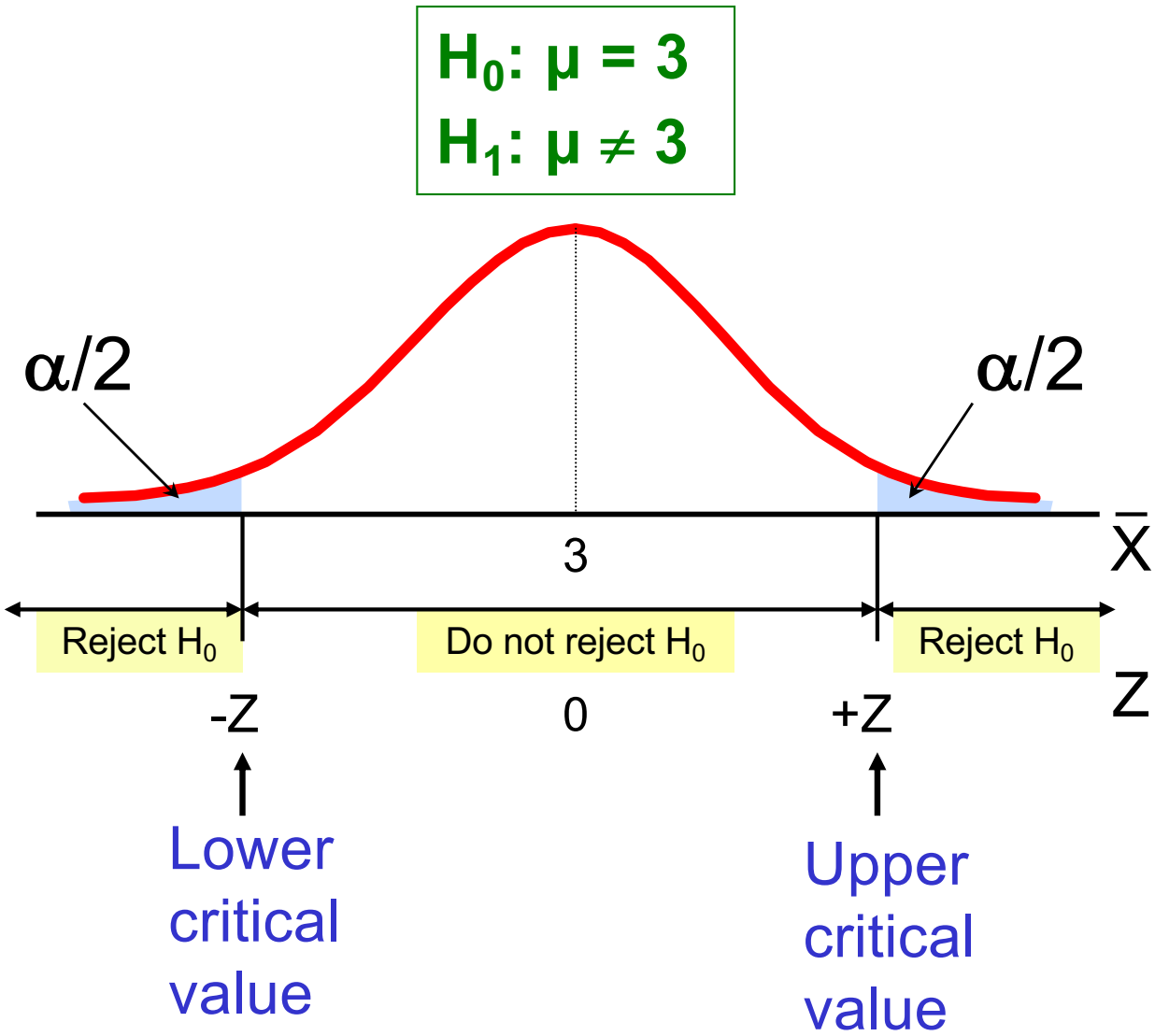
$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Critical Value Approach to Testing

- For a two-tail test for the mean, σ known:
- Convert sample statistic ($\bar{X}$) to test statistic (Z statistic)
- Determine the critical Z values for a specified level of significance α from a table or computer
- Decision Rule: If the test statistic falls in the rejection region, reject H_0 ; otherwise do not reject H_0

Two-Tail Tests

- There are two cutoff values (critical values), defining the regions of rejection



6 Steps in Hypothesis Testing

1. State the null hypothesis, H_0 and the alternative hypothesis, H_1
2. Choose the level of significance, α, and the sample size, n
3. Determine the appropriate test statistic and sampling distribution
4. Determine the critical values that divide the rejection and nonrejection regions

6 Steps in Hypothesis Testing

(continued)

5. Collect data and compute the value of the test statistic
6. Make the statistical decision and state the managerial conclusion. If the test statistic falls into the nonrejection region, do not reject the null hypothesis H_0. If the test statistic falls into the rejection region, reject the null hypothesis. Express the managerial conclusion in the context of the problem

Hypothesis Testing Example

Test the claim that the true mean # of TV sets in US homes is equal to 3. (Assume $\sigma = 0.8$)

1. State the appropriate null and alternative hypotheses
   - H_0: $\mu = 3$ H_1: $\mu \neq 3$ (This is a two-tail test)
2. Specify the desired level of significance and the sample size
   - Suppose that $\alpha = 0.05$ and $n = 100$ are chosen for this test

Hypothesis Testing Example

(continued)

3. Determine the appropriate technique
   - σ is known so this is a Z test.
4. Determine the critical values
   - For $\alpha = 0.05$ the critical Z values are ±1.96
5. Collect the data and compute the test statistic
   - Suppose the sample results are

     $n = 100$, $\bar{X} = 2.84$ ($\sigma = 0.8$ is assumed known)

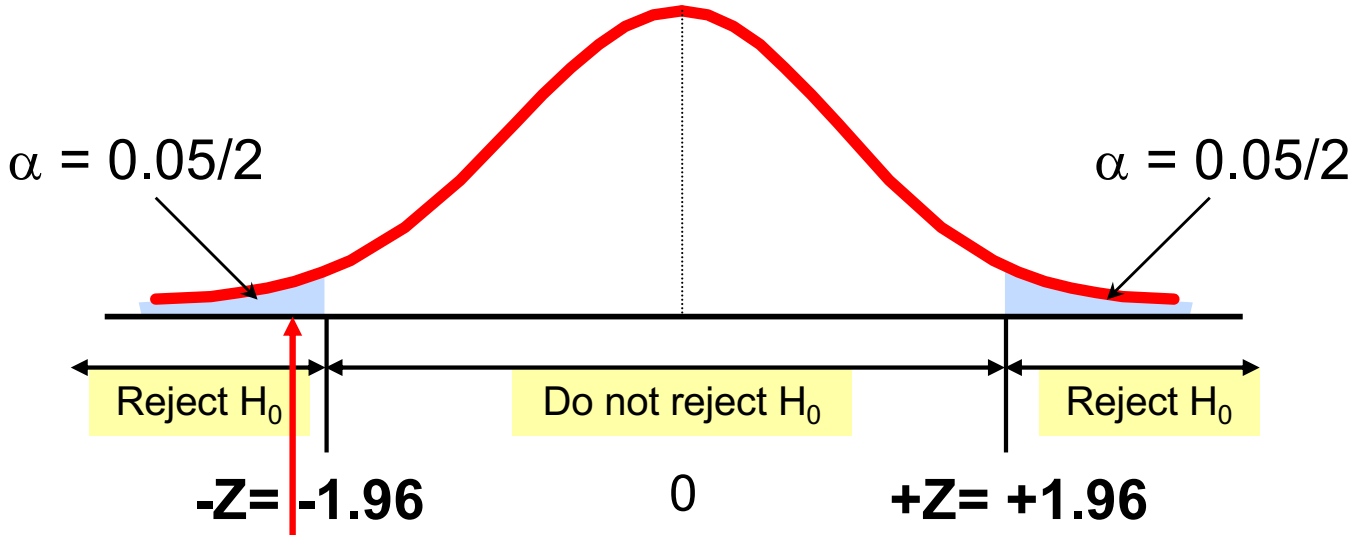
So the test statistic is:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{2.84 - 3}{\frac{0.8}{\sqrt{100}}} = \frac{-.16}{.08} = -2.0$$

Hypothesis Testing Example

(continued)

- 6. Is the test statistic in the rejection region?



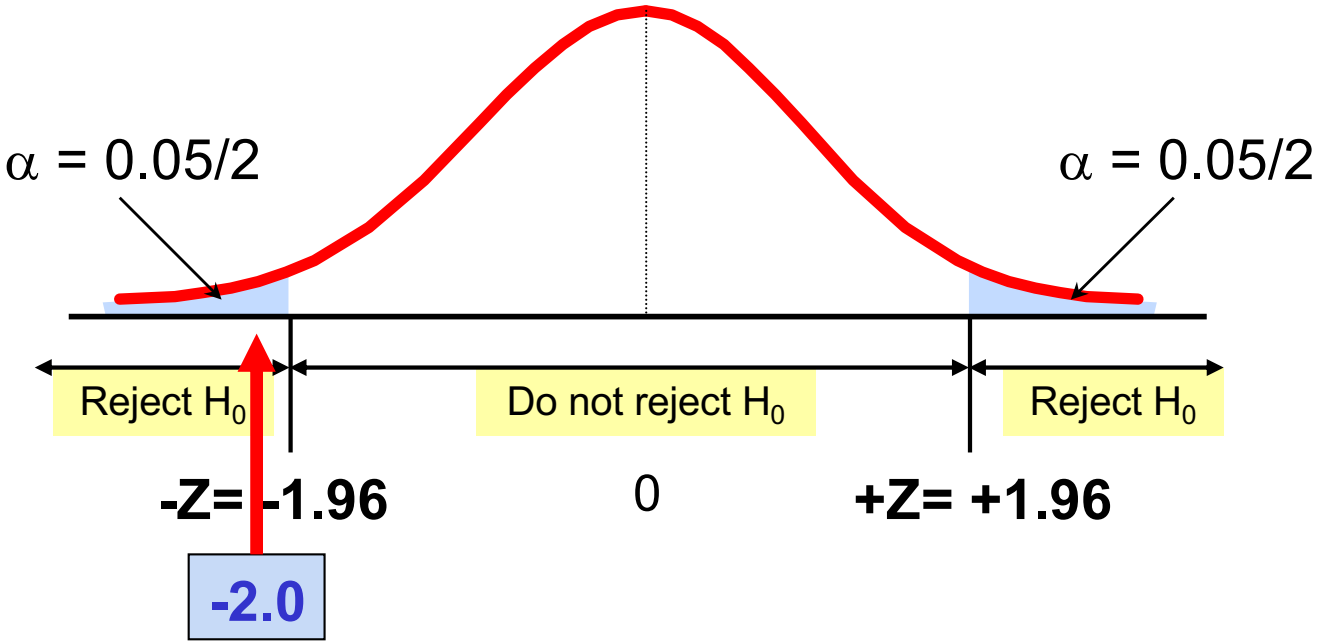
Reject H_0 if $Z < -1.96$ or $Z > 1.96$; otherwise do not reject H_0

Here, $Z = -2.0 < -1.96$, so the test statistic is in the rejection region

Hypothesis Testing Example

(continued)

6(continued). Reach a decision and interpret the result



Since Z = -2.0 < -1.96, we reject the null hypothesis and conclude that there is sufficient evidence that the mean number of TVs in US homes is not equal to 3

p-Value Approach to Testing

- p-value: Probability of obtaining a test statistic more extreme ($\leq$ or $\geq$) than the observed sample value given H_0 is true
  - Also called observed level of significance
  - Smallest value of α for which H_0 can be rejected

p-Value Approach to Testing

(continued)

- Convert Sample Statistic (e.g., $\bar{X}$) to Test Statistic (e.g., Z statistic)
- Obtain the p-value from a table or computer
- Compare the p-value with α

  - If p-value $<$ α , reject H_0
  - If p-value $\geq$ α , do not reject H_0

p-Value Example

- Example: How likely is it to see a sample mean of 2.84 (or something further from the mean, in either direction) if the true mean is μ = 3.0?

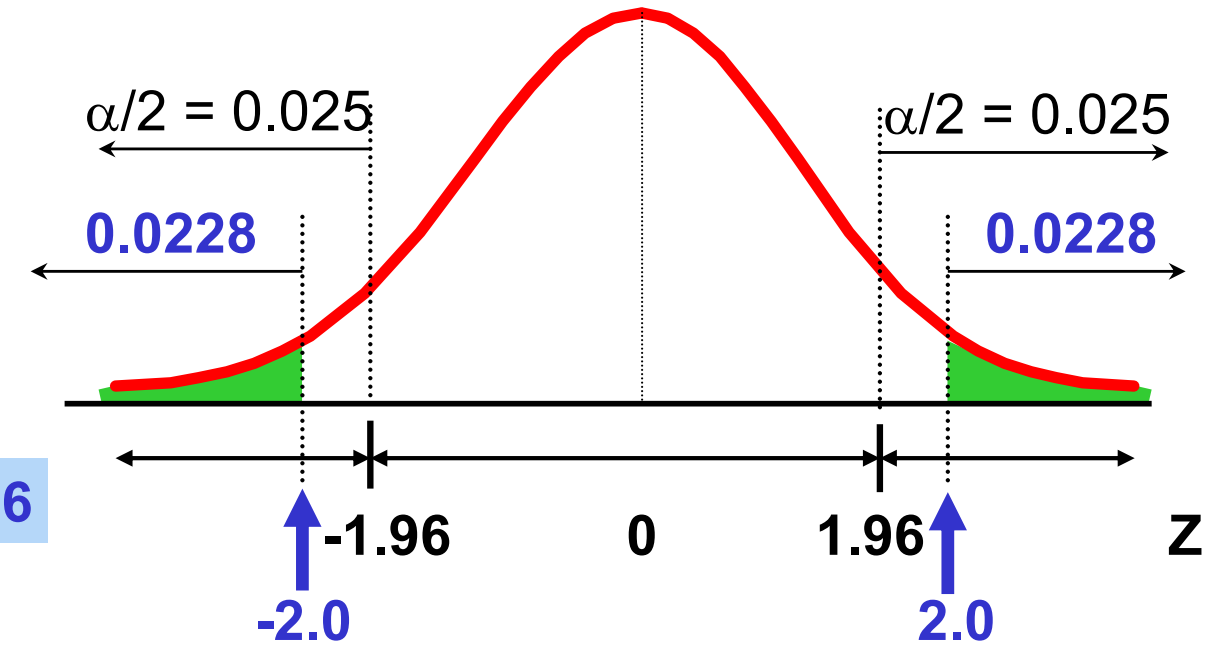
$\overline{X}$ = 2.84 is translated to a Z score of Z = -2.0

$$P(Z < -2.0) = 0.0228$$

$$P(Z > 2.0) = 0.0228$$

p-value

= 0.0228 + 0.0228 = 0.0456



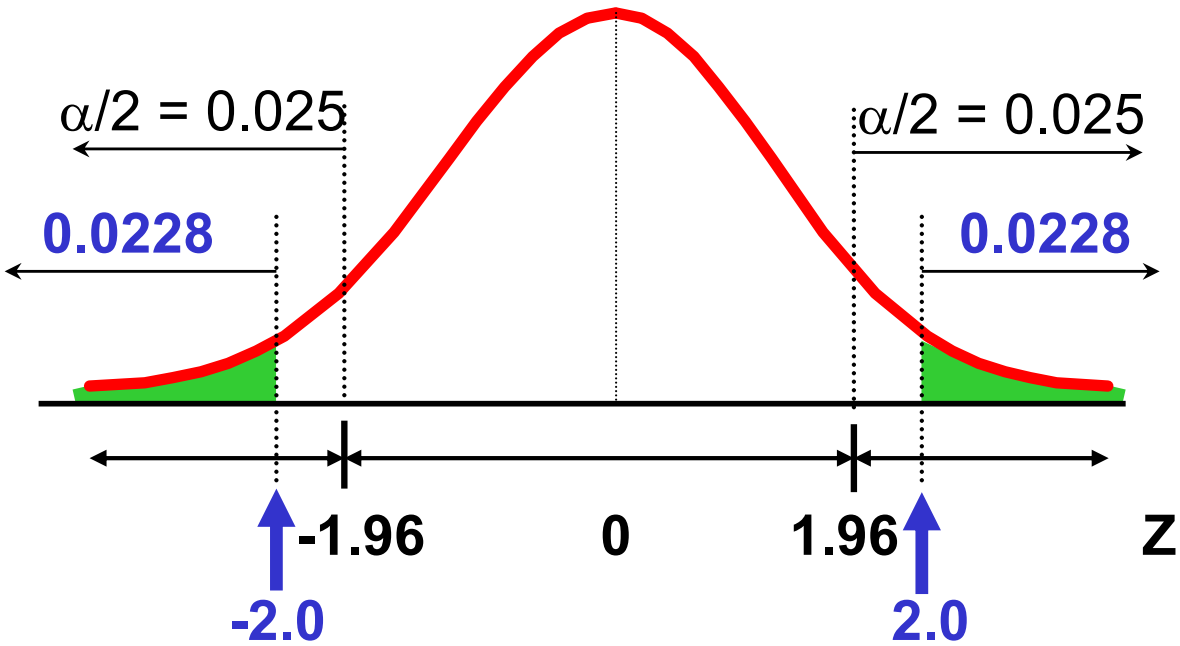
p-Value Example

(continued)

- Compare the p-value with α
  - If p-value $<$ α , reject H_0
  - If p-value $\geq$ α , do not reject H_0

Here: p-value = 0.0456
α = 0.05

Since 0.0456 < 0.05, we reject the null hypothesis



Connection to Confidence Intervals

- For $\overline{X} = 2.84$, $\sigma = 0.8$ and $n = 100$, the 95% confidence interval is:

$$2.84 - (1.96)\frac{0.8}{\sqrt{100}} \quad \text{to} \quad 2.84 + (1.96)\frac{0.8}{\sqrt{100}}$$

$$2.6832 \le \mu \le 2.9968$$

- Since this interval does not contain the hypothesized mean (3.0), we reject the null hypothesis at $\alpha = 0.05$

One-Tail Tests

- In many cases, the alternative hypothesis focuses on a particular direction

$H_0: \mu \geq 3$
$H_1: \mu < 3$

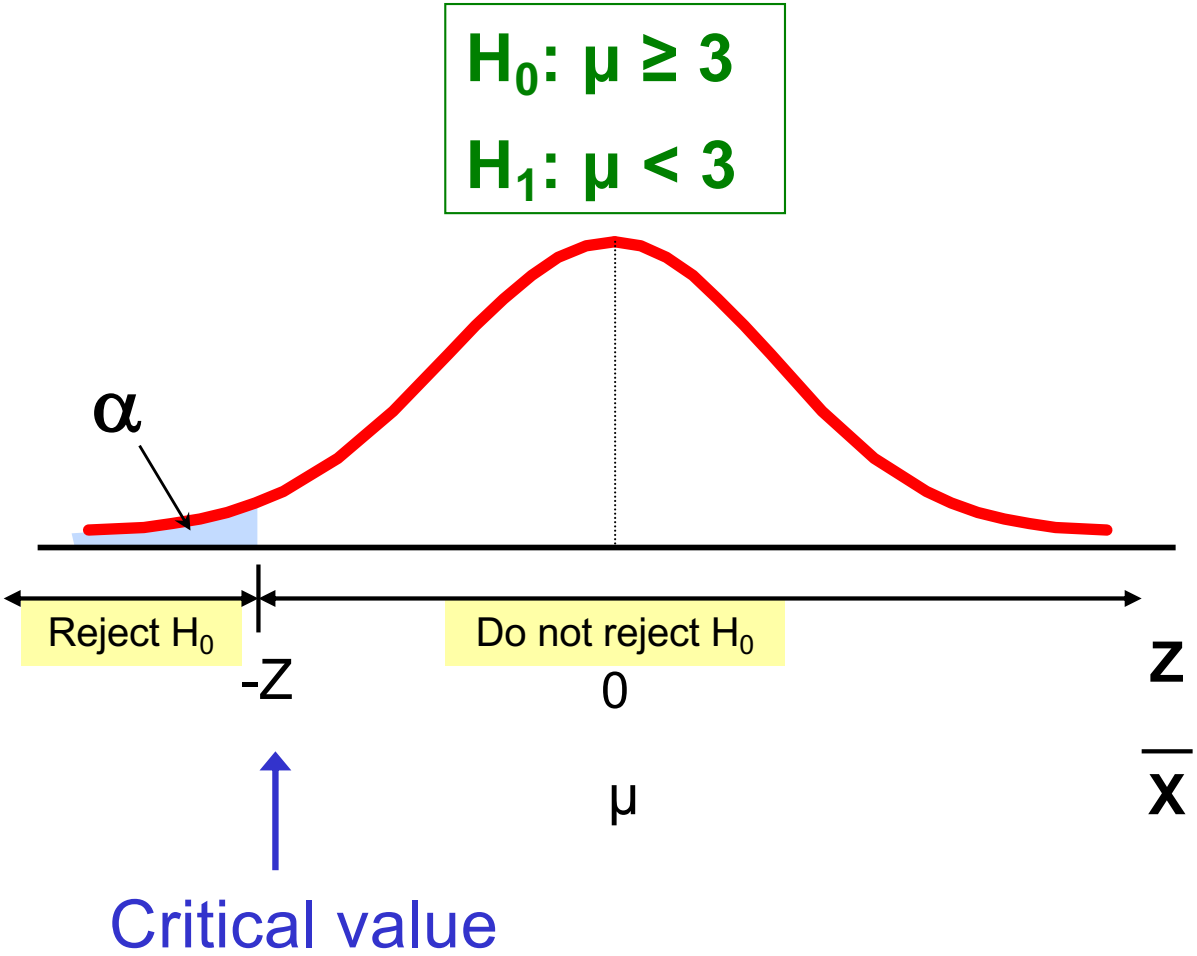
This is a lower-tail test since the alternative hypothesis is focused on the lower tail below the mean of 3

$H_0: \mu \leq 3$
$H_1: \mu > 3$

This is an upper-tail test since the alternative hypothesis is focused on the upper tail above the mean of 3

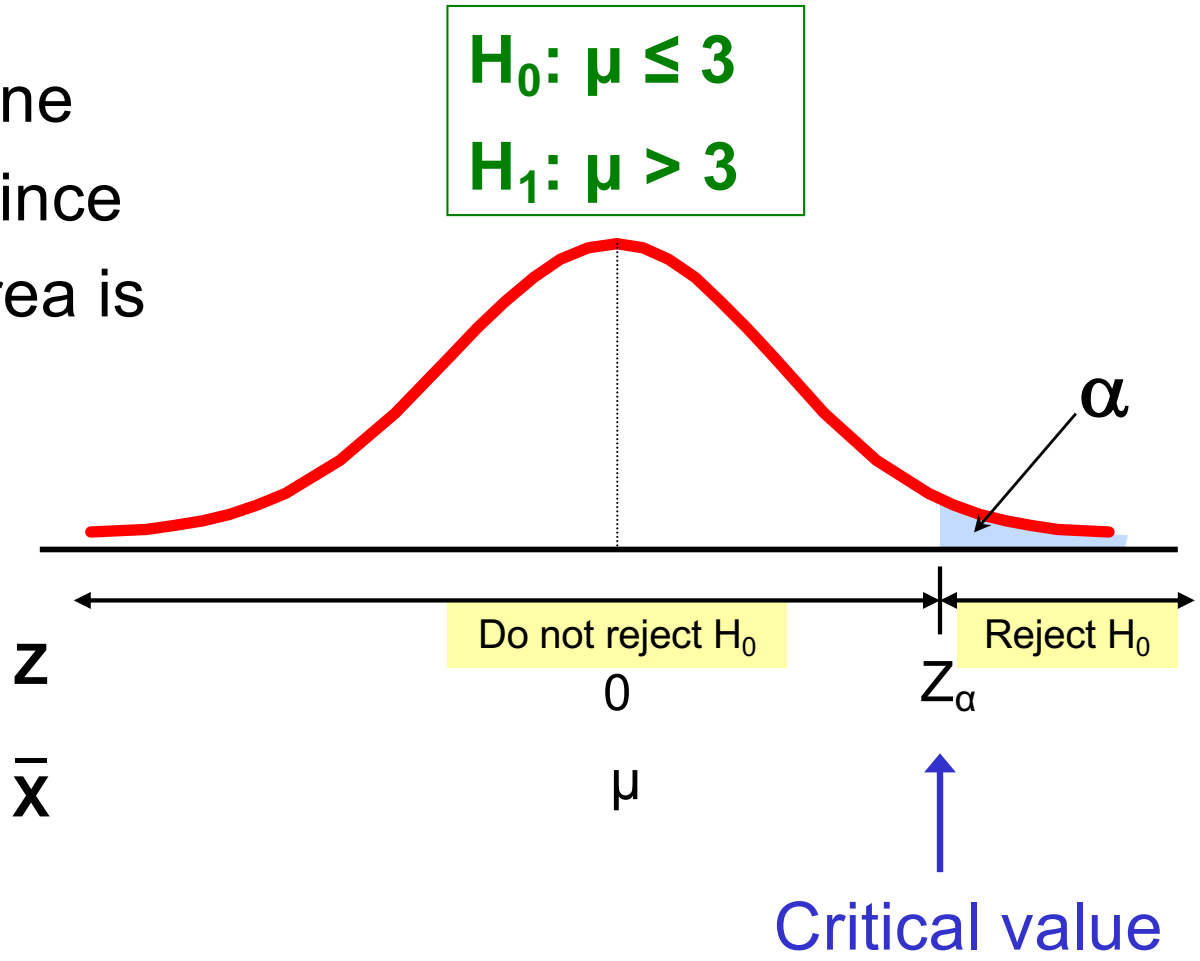
Lower-Tail Tests

- There is only one critical value, since the rejection area is in only one tail



Upper-Tail Tests

- There is only one critical value, since the rejection area is in only one tail



Example: Upper-Tail Z Test for Mean (σ Known)

A phone industry manager thinks that customer monthly cell phone bills have increased, and now average over \$52 per month. The company wishes to test this claim. (Assume σ = 10 is known)

Form hypothesis test:

H_0: $\mu \leq 52$ the average is not over \$52 per month

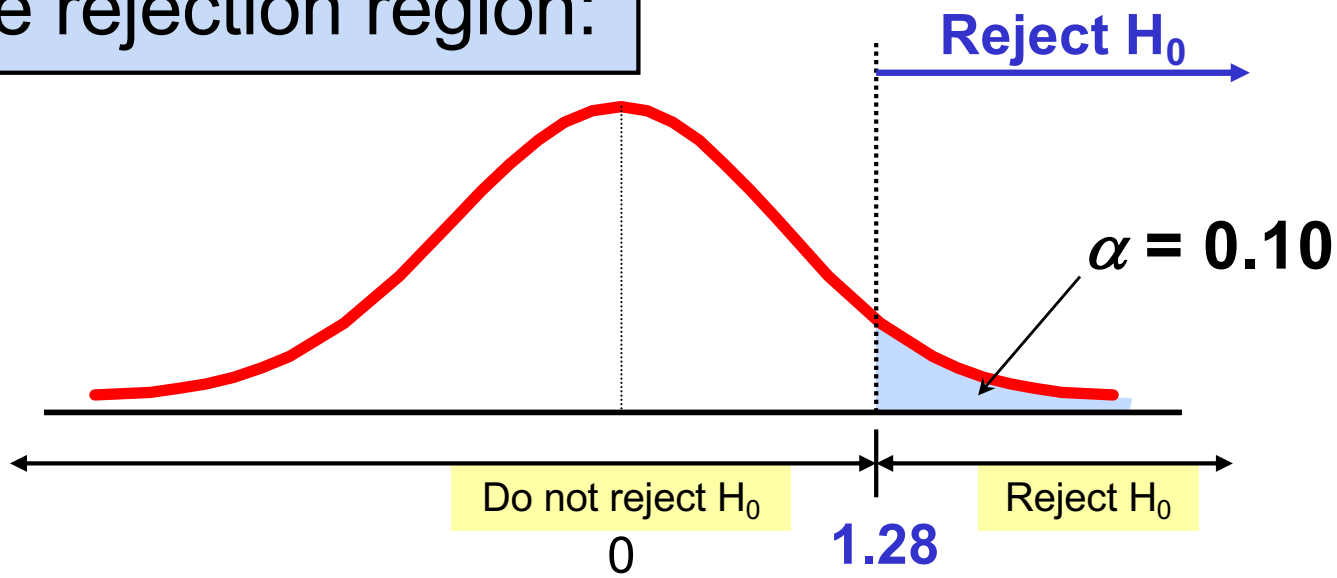
H_1: $\mu > 52$ the average **is** greater than \$52 per month
(i.e., sufficient evidence exists to support the manager's claim)

Example: Find Rejection Region

(continued)

- Suppose that α = 0.10 is chosen for this test

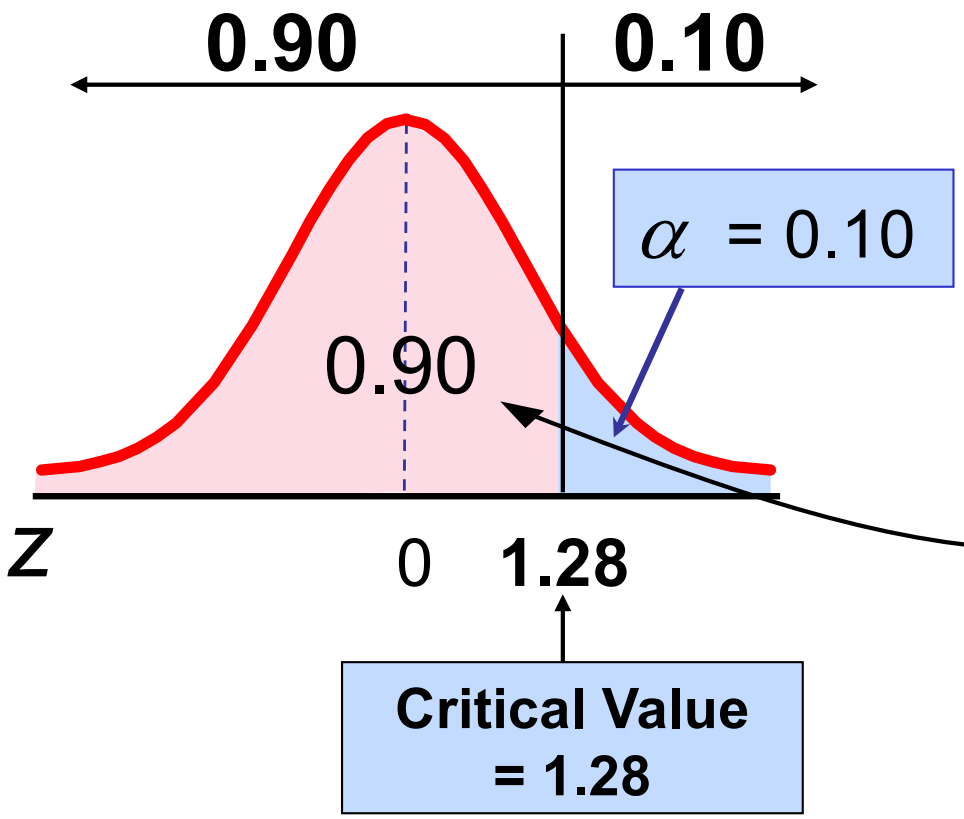
Find the rejection region:



Reject H_0 if $Z > 1.28$

Review: One-Tail Critical Value

What is Z given α = 0.10?



Standardized Normal Distribution Table (Portion)

Z	.07	.08	.09
1.1	.8790	.8810	.8830
1.2	.8980	.8997	.9015
1.3	.9147	.9162	.9177

Example: Test Statistic

(continued)

Obtain sample and compute the test statistic

Suppose a sample is taken with the following results: $n = 64$, $\bar{X} = 53.1$ (σ=10 was assumed known)

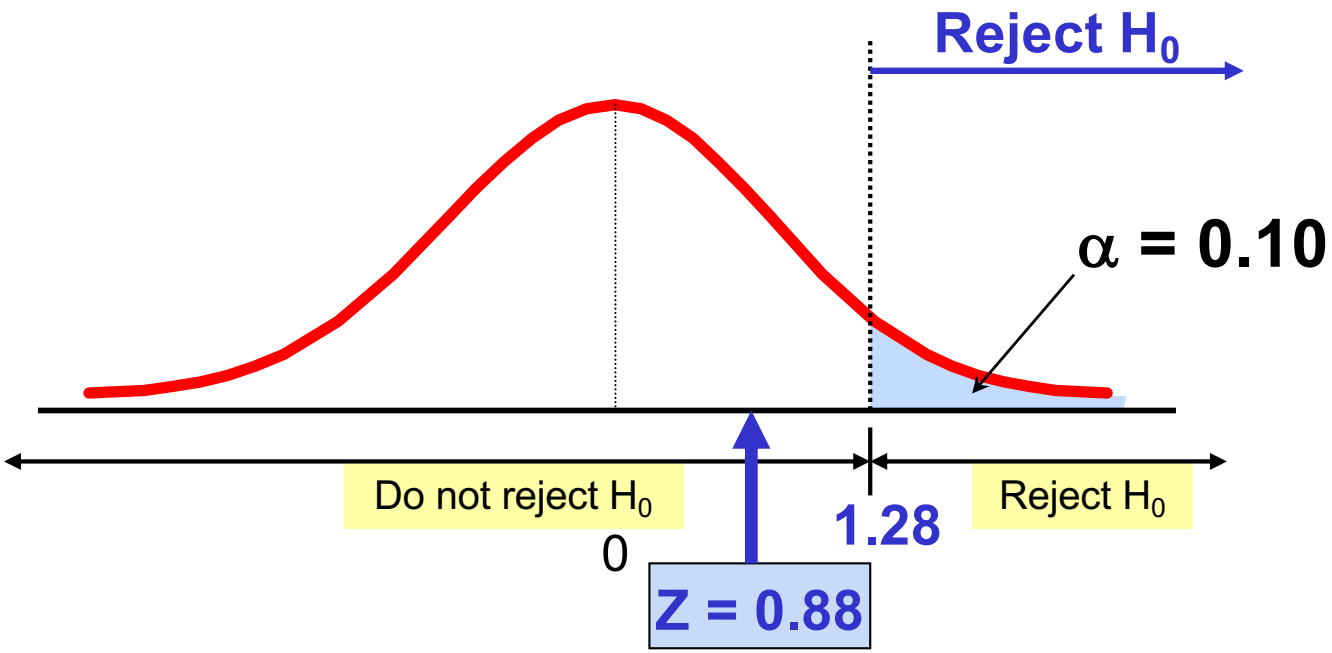
- Then the test statistic is:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{53.1 - 52}{\frac{10}{\sqrt{64}}} = 0.88$$

Example: Decision

(continued)

Reach a decision and interpret the result:



Do not reject H_0 since $Z = 0.88 \leq 1.28$

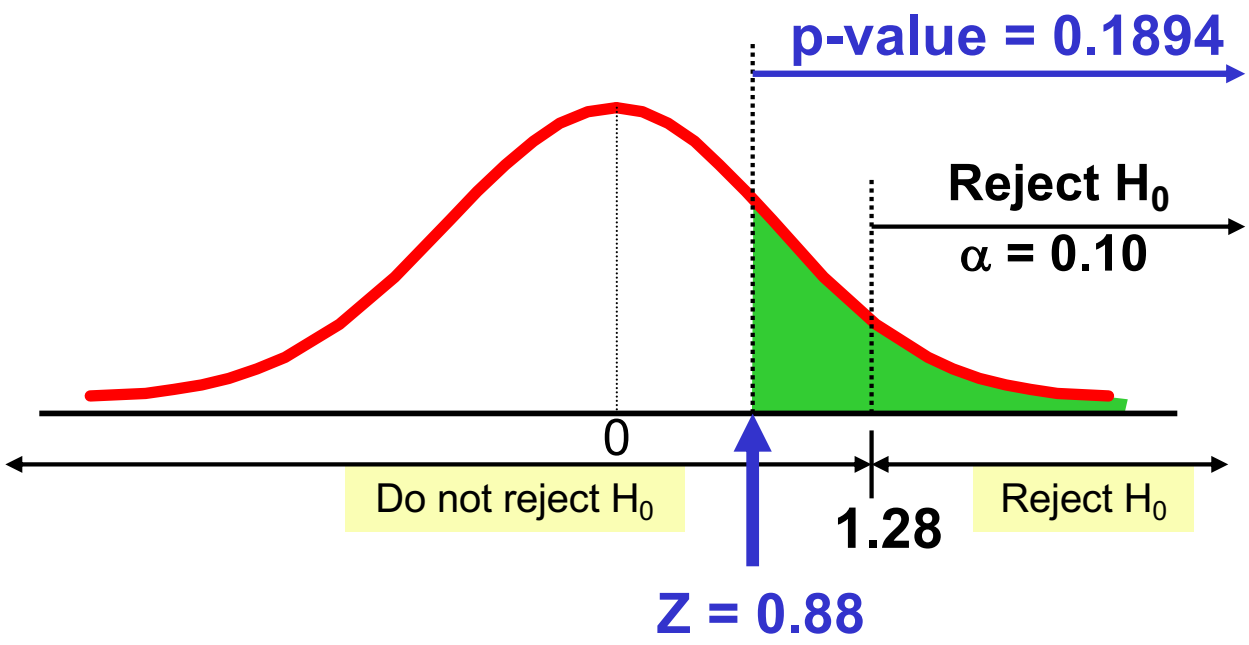
i.e.: there is not sufficient evidence that the mean bill is over $52

p -Value Solution

(continued)

Calculate the p-value and compare to α

(assuming that μ = 52.0)



$$P(\bar{X} \geq 53.1)$$

$$= P\left(Z \geq \frac{53.1 - 52.0}{10/\sqrt{64}}\right)$$

$$= P(Z \geq 0.88) = 1 - 0.8106$$

$$= 0.1894$$

Do not reject H_0 since p-value = 0.1894 > α = 0.10

Class Exercise

- A company that manufactures chocolate bars is particularly concerned that the mean weight of the chocolate bar not exceed 6.03 grams. Past experience allows you to assume that the standard deviation is 0.02 grams. A sample of 50 chocolate bars is selected and the sample mean is 6.034 grams. Using the α=0.01 level of significance, is there evidence that the population mean weight of the chocolate bars is greater than 6.03 grams?
- Use critical value and p-value approaches

Class Exercise

- Use critical value approach

Class Exercise

- Use p-value approach

t Test of Hypothesis for the Mean (σ Unknown)

- Convert sample statistic ($\bar{X}$) to a t test statistic

Hypothesis Tests for μ

- **σ Known (Z test)**
- **σ Unknown (t test)**

The test statistic is:

$$t_{n-1} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$$

Example: Two-Tail Test (σ Unknown)

The average cost of a hotel room in New York is said to be \$168 per night. A random sample of 25 hotels resulted in $\bar{X}$ = \$172.50 and S = \$15.40. Test at the $\alpha = 0.05$ level.

(Assume the population distribution is normal)

$H_0: \mu = 168$

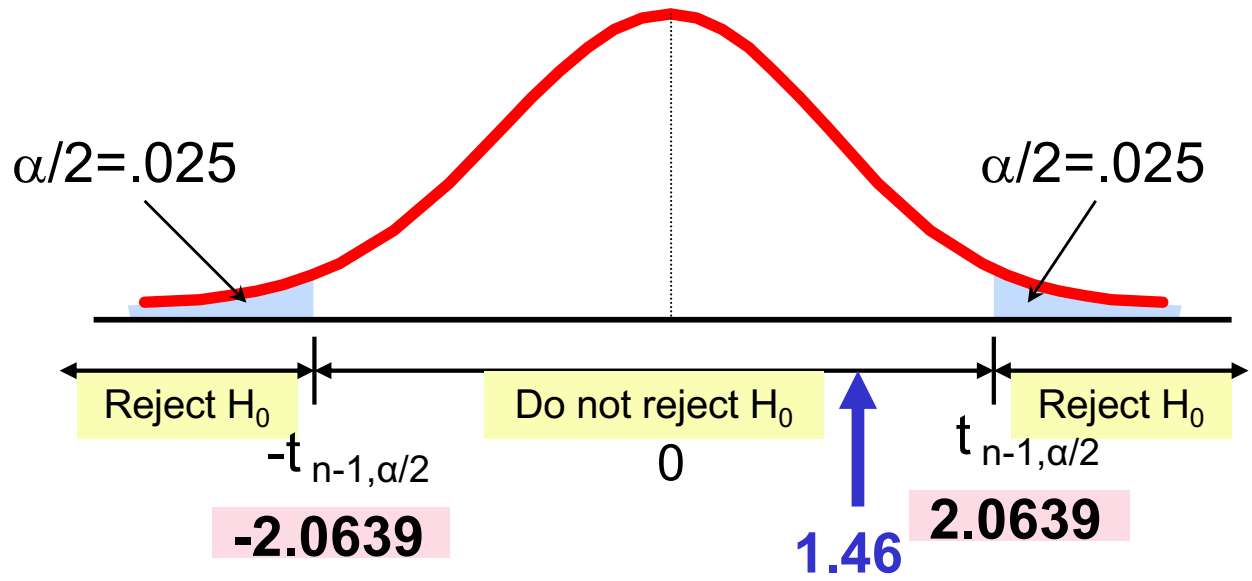
$H_1: \mu \neq 168$

Example Solution: Two-Tail Test

$H_0: \mu = 168$
$H_1: \mu \neq 168$

- $\alpha = 0.05$
- $n = 25$
- σ is unknown, so use a **t statistic**
- **Critical Value:**

$t_{24} = \pm 2.0639$



$$t_{n-1} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{172.50 - 168}{\frac{15.40}{\sqrt{25}}} = 1.46$$

Do not reject H_0: not sufficient evidence that true mean cost is different than \$168

Connection to Confidence Intervals

- For $\overline{X}$ = 172.5, S = 15.40 and n = 25, the 95% confidence interval is:

$$172.5 - (2.0639)\ 15.4/\sqrt{25} \quad \text{to} \quad 172.5 + (2.0639)\ 15.4/\sqrt{25}$$

$$166.14 \le \mu \le 178.86$$

- Since this interval contains the Hypothesized mean (168), we do not reject the null hypothesis at $\alpha = 0.05$

Hypothesis Tests for Proportions

- Involves categorical variables
- Two possible outcomes
  - "Success" (possesses a certain characteristic)
  - "Failure" (does not possess that characteristic)
- Fraction or proportion of the population in the "success" category is denoted by π

Proportions

(continued)

- Sample proportion in the success category is denoted by p
  - $$p = \frac{X}{n} = \frac{\text{number of successes in sample}}{\text{sample size}}$$
- When both $n\pi$ and $n(1-\pi)$ are at least 5, p can be approximated by a normal distribution with mean and standard deviation
  - $$\mu_p = \pi \qquad \sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

Hypothesis Tests for Proportions

- The sampling distribution of p is approximately normal, so the test statistic is a Z value:

$$Z = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Hypothesis Tests for p

- **$n\pi \geq 5$ and $n(1-\pi) \geq 5$**
- **$n\pi < 5$ or $n(1-\pi) < 5$**
  - Not discussed in this chapter

Z Test for Proportion in Terms of Number of Successes

- An equivalent form to the last slide, but in terms of the number of successes, X:

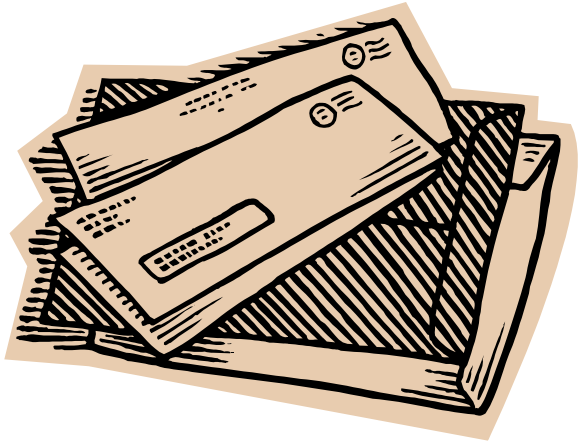
$$Z = \frac{X - n\pi}{\sqrt{n\pi(1-\pi)}}$$

Hypothesis Tests for X

- **$X \geq 5$ and $n\text{-}X \geq 5$**
- **$X < 5$ or $n\text{-}X < 5$**
  - Not discussed in this chapter

Example: Z Test for Proportion

A marketing company claims that it receives 8% responses from its mailing. To test this claim, a random sample of 500 were surveyed with 25 responses. Test at the $\alpha = 0.05$ significance level.



Check:

$n\pi = (500)(.08) = 40$

$n(1-\pi) = (500)(.92) = 460$

✓

Z Test for Proportion: Solution

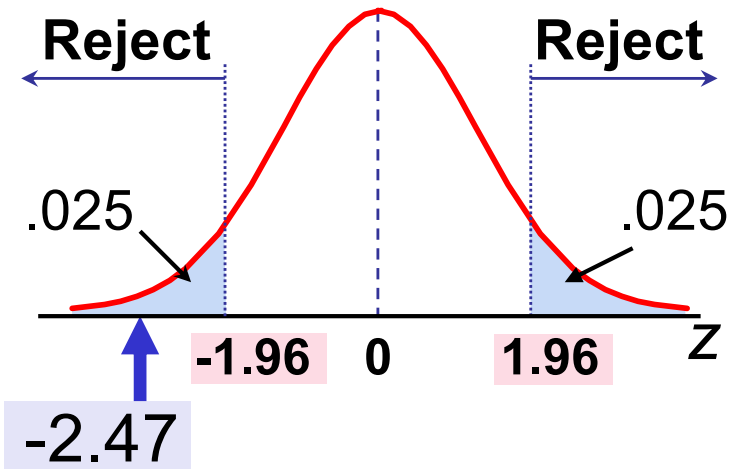
$H_0: \pi = 0.08$

$H_1: \pi \neq 0.08$

$\alpha = 0.05$

$n = 500, \quad p = 0.05$

Critical Values: ± 1.96



Test Statistic:

$$Z = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{.05 - .08}{\sqrt{\frac{.08(1-.08)}{500}}} = -2.47$$

Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

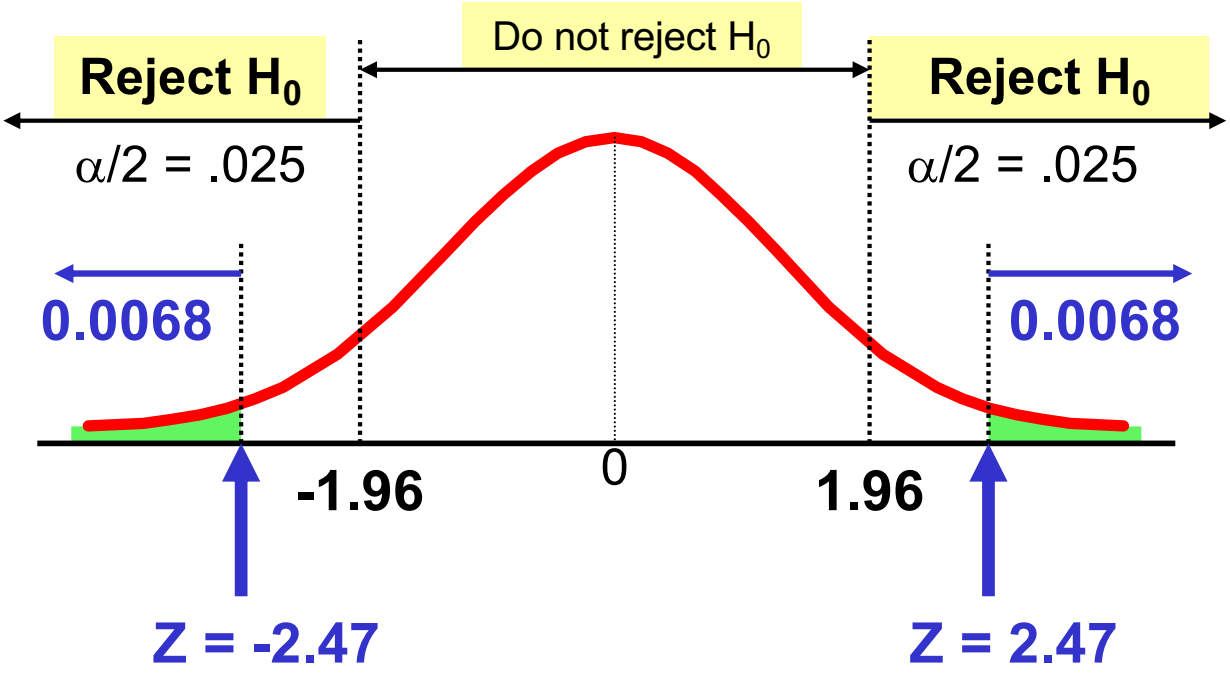
There is sufficient evidence to reject the company's claim of 8% response rate.

p-Value Solution

(continued)

Calculate the p-value and compare to α

(For a two-tail test the p-value is always two-tail)



p-value = 0.0136:

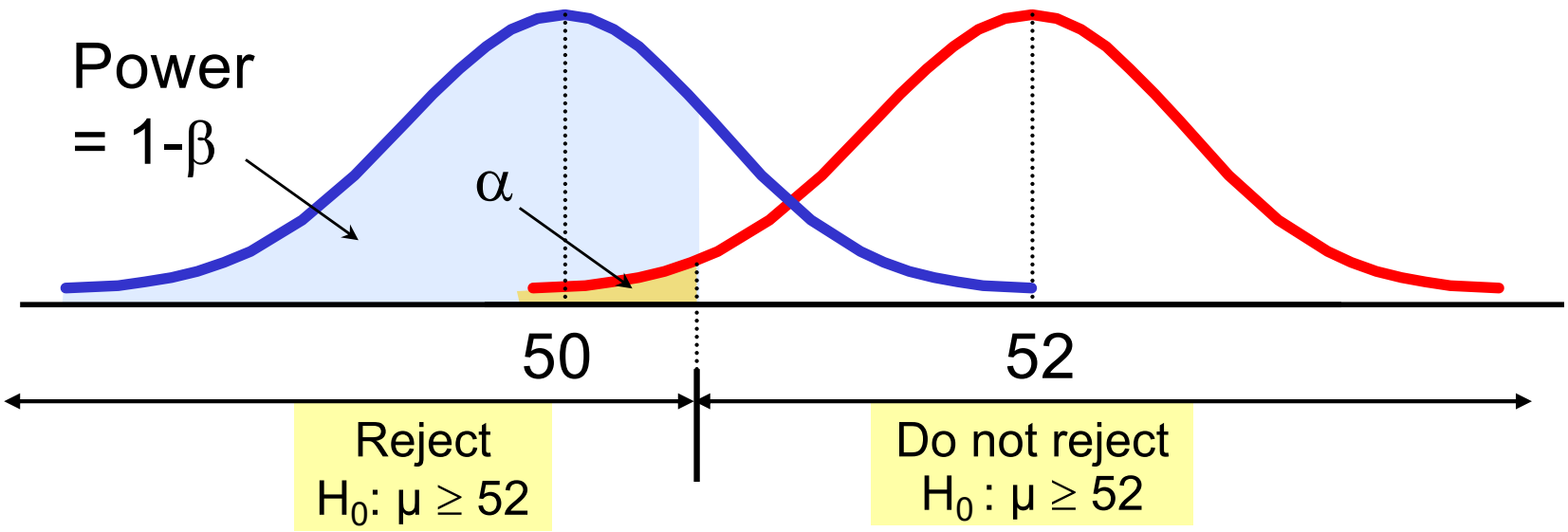
$$P(Z \le -2.47) + P(Z \ge 2.47)$$
$$= 2(0.0068) = 0.0136$$

Reject H_0 since p-value = 0.0136 < α = 0.05

The Power of a Test

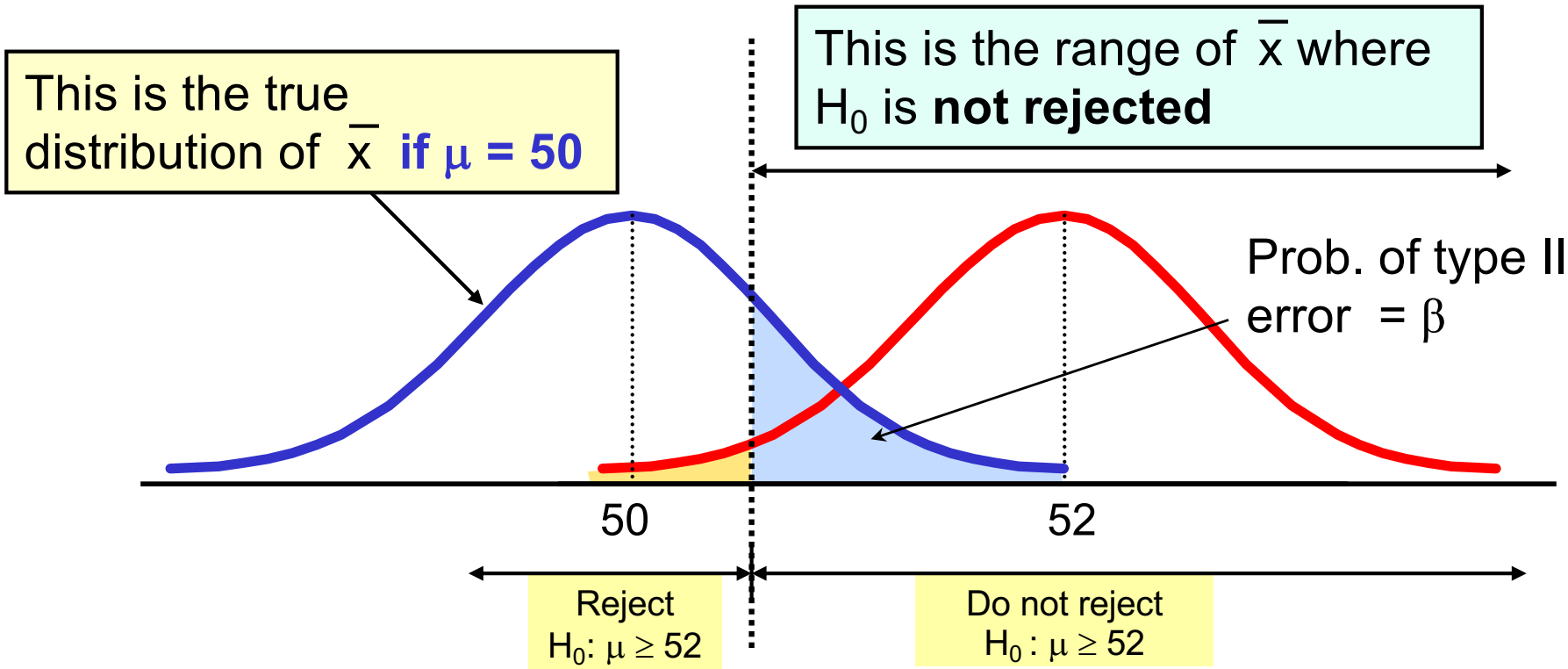
- **The power of the test is the probability of correctly rejecting a false H_0**

Suppose we correctly reject **H_0: $\mu \geq 52$** when in fact the true mean is **$\mu = 50$**



Type II Error

- Suppose we do not reject H_0: $\mu \geq 52$ when in fact the true mean is $\mu = 50$

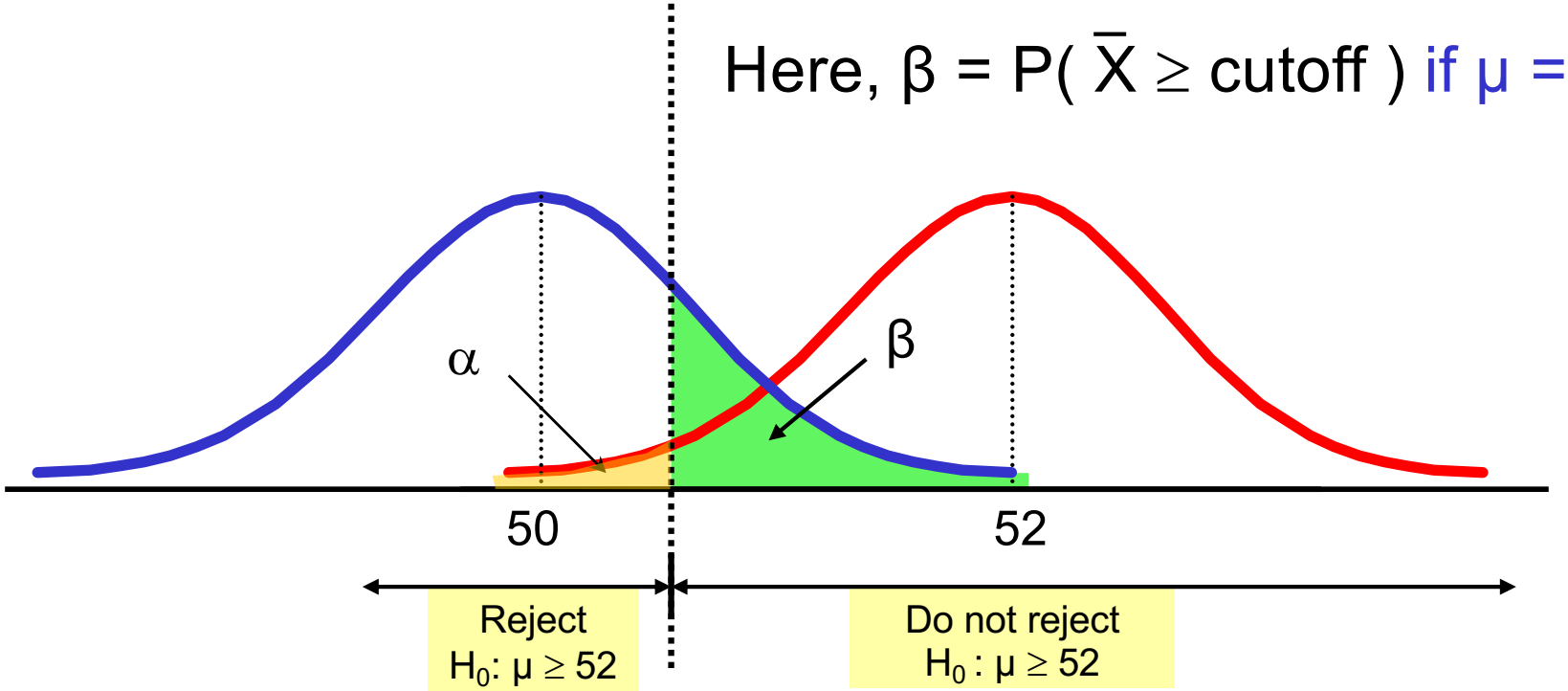


Type II Error

(continued)

- Suppose we do not reject H_0: $\mu \geq 52$ when in fact the true mean is $\mu = 50$

Here, $\beta = P(\bar{X} \geq \text{cutoff})$ if $\mu = 50$



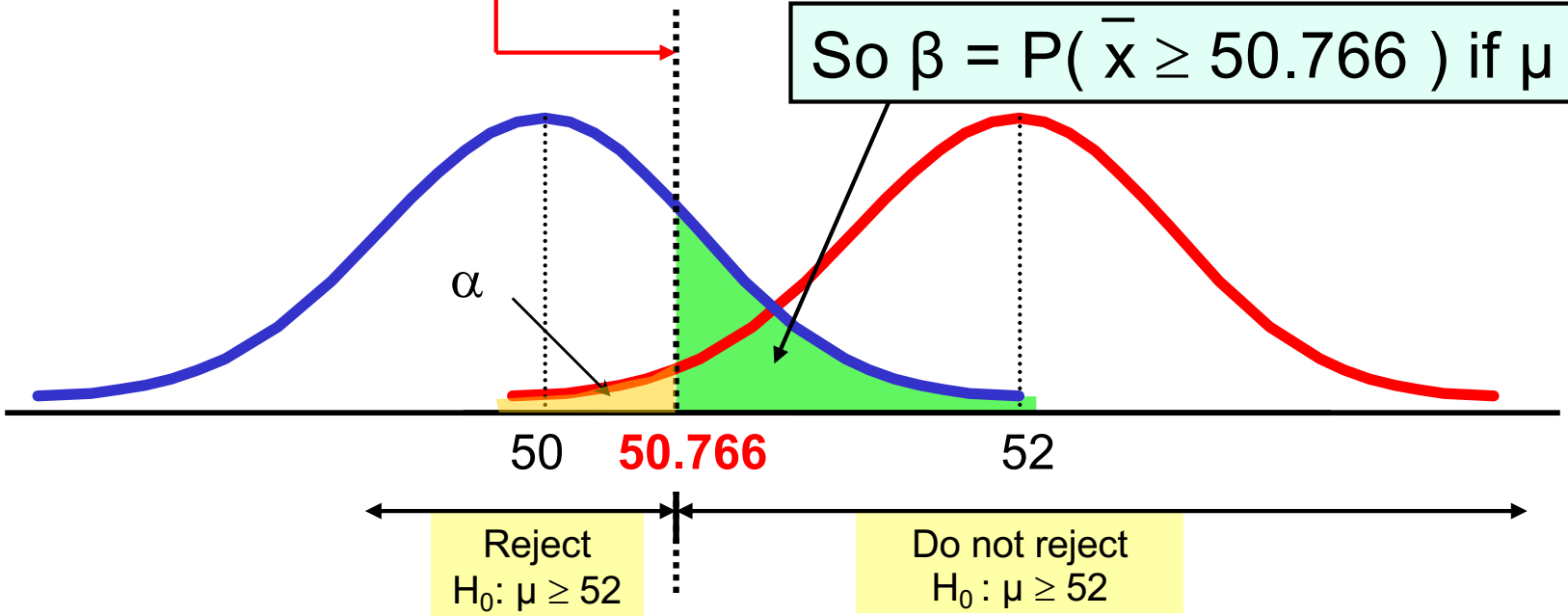
Calculating β

- Suppose n = 64 , σ = 6 , and α = .05

$$\text{cutoff} = \bar{X}_\alpha = \mu - Z_\alpha \frac{\sigma}{\sqrt{n}} = 52 - 1.645 \frac{6}{\sqrt{64}} = \boxed{50.766}$$

(for $H_0 : \mu \geq 52$)

So $\beta = P(\bar{x} \geq 50.766)$ if $\mu = 50$



Calculating β and Power of the test

(continued)

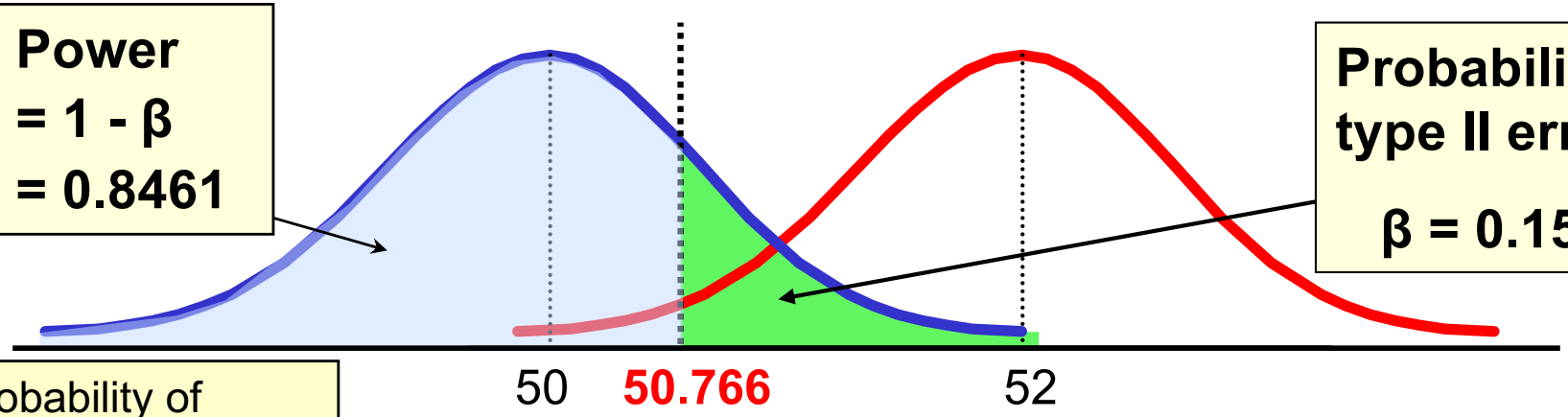
- Suppose n = 64 , σ = 6 , and α = 0.05

$$P(\bar{x} \geq 50.766 \mid \mu = 50) = P\left(z \geq \frac{50.766 - 50}{6/\sqrt{64}} \right) = P(z \geq 1.02) = 1.0 - 0.8461 = \boxed{0.1539}$$

Power
= 1 - β
= 0.8461

Probability of type II error:

β = 0.1539



The probability of correctly rejecting a false null hypothesis is 0.8641

Power of the Test

- Conclusions regarding the power of the test:

  1. A one-tail test is more powerful than a two-tail test
  2. An increase in the level of significance (α) results in an increase in power
  3. An increase in the sample size results in an increase in power

Potential Pitfalls and Ethical Considerations

- Use randomly collected data to reduce selection biases
- Do not use human subjects without informed consent
- Choose the level of significance, α, and the type of test (one-tail or two-tail) before data collection
- Do not employ "data snooping" to choose between one-tail and two-tail test, or to determine the level of significance

  Data snooping = Using the same data to both discover and validate a finding → leads to biased, unreliable results.
- Do not practice "data cleansing" to hide observations that do not support a stated hypothesis
- Report all pertinent findings

  relevant

Chapter Summary

- Addressed hypothesis testing methodology
- Performed Z Test for the mean (σ known)
- Discussed critical value and p–value approaches to hypothesis testing
- Performed one-tail and two-tail tests

Chapter Summary

(continued)

- Performed t test for the mean (σ unknown)
- Performed Z test for the proportion
- Calculated the probability of a Type II error and the power of the test
- Discussed pitfalls and ethical issues